CYBERSECURITY LAB REPORT

GoPhish Security-Awareness Lab & Findings Report

IG-AWARENESS-01 | Controlled Local Simulation

Technical report documenting the setup, execution, observed results, evidence, and limitations of a localhost-only GoPhish simulation using MailHog.

PROJECT PROFILE	DETAIL
Project type	Hands-on cybersecurity / security awareness
Primary platform	GoPhish
Supporting service	MailHog for local email capture
Environment	Kali Linux; localhost-only services
Campaign	IG-AWARENESS-01 HTML Verification
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Date	23 October 2026

Report scope

This report records a controlled local test. It does not describe a real-world phishing incident, compromise, or independent user failure.

1. Executive Summary

I configured GoPhish to run a controlled security-awareness campaign against one synthetic recipient inside a local environment. MailHog provided a safe place to inspect captured messages without sending the exercise to a real external inbox. The campaign used an HTML email about an unusual account-security notification and a custom educational landing page that explains warning signs and safer responses.

The captured campaign dashboard showed one email sent, one email opened, one link click, zero submitted data, and zero reports. The exported campaign events CSV visibly records campaign creation, email sent, and a link click. The raw events export shown in the evidence does not visibly contain an open event, so the dashboard's open count and the raw event export should be treated as a reporting discrepancy to note, not silently reconciled. The click event's recorded IP is 127.0.0.1, consistent with local testing.

Project outcomes

- Configured a GoPhish sending profile using MailHog's local SMTP listener at 127.0.0.1:1025.
- Created an HTML security-awareness email and an educational landing page.
- Used a synthetic recipient (analyst-test@example.test) rather than a real external target.
- Reviewed the campaign dashboard and exported both event and results CSV files.
- Kept the exercise local and designed the landing page not to request or collect account credentials.

2. Test Objective and Success Criteria

Phishing messages can imitate familiar branding, create urgency, and direct recipients toward unverified links. The objective of this project was to build a small, repeatable environment for observing the GoPhish campaign workflow and presenting an educational message that teaches people to pause, inspect, verify independently, and report suspicious messages.

Success criteria

- GoPhish can be configured with a local SMTP destination.
- The synthetic recipient can receive a campaign message in MailHog.
- The campaign's local landing page renders the awareness content.
- Campaign status and CSV exports can be inspected and documented.
- The project avoids real external recipients and does not collect passwords or authentication codes.

3. Test Environment and Architecture

Component	Role	Local endpoint / note
Kali Linux	Host environment for the tools	Local virtual/lab environment
GoPhish admin server	Campaign and template management	https://127.0.0.1:3333
GoPhish phishing server	Serves the campaign landing page	http://127.0.0.1:8080
MailHog SMTP	Captures outgoing test messages	127.0.0.1:1025
MailHog web UI	Inspects captured email	http://127.0.0.1:8025

All service addresses above use loopback (127.0.0.1), so they refer to the local machine. The .test recipient domain is intended for testing, not ordinary public email delivery.

4. Test Procedure and Configuration

4.1 Campaign assets

Asset	Configuration / purpose
Email template	IG-AWARENESS-01 Security Alert; subject: “Security notice: Review an unusual sign-in”
Sending profile	IG-AWARENESS-01 MailHog SMTP; host 127.0.0.1:1025; no SMTP credentials shown/configured
Recipient group	IG-AWARENESS-01 Synthetic Users; one synthetic recipient, analyst-test@example.test
Landing page	IG-AWARENESS-01 Awareness Training; explains warning signs and safe response
Campaign	IG-AWARENESS-01 HTML Verification

4.2 Landing-page content

The landing page is an educational debrief, not a credential-harvesting page. Its content covers unexpected urgency, unfamiliar senders, unverified links, requests for secrets, and unusual context. It advises users to open the official app or type a known website address independently, never share passwords or one-time codes, and report suspicious messages.

4.3 Workflow

1. Create and review the HTML email template.
2. Configure the MailHog SMTP sending profile.
3. Create a group containing a synthetic test recipient.
4. Configure the campaign with the local landing-page URL.
5. Launch the campaign only within the local test environment.
6. Inspect the delivered campaign email in MailHog and visit the local educational landing page.
7. Review the GoPhish results dashboard and export campaign events and results CSV files.

5. Evidence Register

This register maps each captured screenshot to the configuration or result it supports. Original image files and exported CSVs should be retained alongside this report.

Evidence	Filename	What it supports
01	01 Email Template Preview.png	HTML email template and subject/content preview.
02	02 Sending Profile.png	MailHog SMTP host and sender configuration.
03	03 Test Recipient Group.png	Synthetic recipient and group configuration.
04	04 SMTP Test Email.png	GoPhish's default SMTP test message captured in MailHog; this is a connectivity test, not the campaign email.
05	05 Campaign Setup.png	Campaign configuration, selected assets, local URL, and synthetic group.
06	06 Campaign Results.png	Dashboard counts and status shown for the campaign.
07	07 HTML Email in MailHog.png	Campaign HTML email captured by MailHog.
08	08 Landing Page Preview - Top.png	Top of the local educational landing page.
09	09 Landing Page Preview - Bottom.png	Lower landing-page sections and safety messaging.
10	10 Campaign Raw Events.png	Events CSV opened in a text editor; shows campaign creation, email sent, and link click.
11	11 Campaign Results CSV.png	Results CSV opened in LibreOffice Calc; shows the recipient status as Clicked Link.

6. Findings and Observed Results

Check	Observed evidence	Interpretation
Email delivery to test capture	Campaign email visible in MailHog.	Pass: local capture confirms the campaign message reached the test mailbox service.
HTML email rendering	HTML message displays the security-awareness content in MailHog.	Pass: the designed email content is present.
Landing page	Local page renders the educational content at 127.0.0.1:8080.	Pass: local training page is accessible.
Email sent	Dashboard shows 1; events CSV includes Email Sent.	Pass: send event is recorded.
Email opened	Dashboard shows 1; raw events screenshot does not visibly list an Email Opened event.	Needs qualification: dashboard and event export do not fully agree in the captured evidence.
Link clicked	Dashboard shows 1; events CSV includes Clicked Link; results CSV status is Clicked Link.	Pass: a local click was recorded. Treat as the author's controlled test activity.
Submitted data	Dashboard shows 0.	No submitted-data event is shown.
Email reported	Dashboard shows 0; results CSV reported=false.	No report event is shown.

Important interpretation: the open and click events are from controlled local testing, not proof that an independent person fell for phishing. The campaign event export contains a click payload with address 127.0.0.1 and a local browser user-agent, supporting the interpretation that the click came from the local test machine.

7. Technical Analysis

- Email delivery and campaign tracking are separate checks. MailHog confirms message capture; GoPhish records campaign interaction events.
- A dashboard metric should be checked against the underlying event export. Here, the open count appears in the dashboard but not in the visible raw-events CSV, so the discrepancy should be documented rather than assumed away.
- The localhost address in the click event is consistent with a local browser visit and should not be presented as an external participant.
- The synthetic .test recipient and MailHog reduce the risk of contacting a real person or sending messages onto the public email system.
- An awareness landing page can turn a simulation into a learning moment by explaining warning signs and safe actions after the click.

8. Safety Controls and Scope

- The project is restricted to a local, controlled environment.
- The recipient is synthetic and uses the reserved .test domain.
- MailHog is used to inspect test messages locally rather than deliver them to a real external inbox.
- The landing page is educational and explicitly says it does not request or collect account credentials.
- No passwords, authentication codes, or real account data should be collected.
- Any future test involving real people or organizational systems would require explicit authorization, an agreed scope, and an approved reporting/debrief process.

9. Limitations and Recommended Follow-up

- Investigate the dashboard-versus-raw-events discrepancy for the email-open metric, then capture fresh evidence if the result changes.

- Add a short README with setup steps, architecture, safety boundaries, and a screenshot index.
- If publishing publicly, remove sensitive host details where appropriate and inspect configuration files, databases, logs, and screenshots for secrets or personal information.
- Add a clear diagram of the local message flow: GoPhish → MailHog SMTP → MailHog UI, plus browser → GoPhish landing page.
- For a future version, document how the exercise would be debriefed and how awareness outcomes would be measured without collecting credentials.

10. Conclusion

IG-AWARENESS-01 demonstrates a small end-to-end GoPhish security-awareness workflow: campaign assets were configured, a message was captured in MailHog, an educational landing page was served locally, and campaign results were exported for review. The evidence supports a controlled local simulation, with one click recorded and no submitted data. The dashboard's open count should remain qualified because the captured raw-events export does not visibly show a matching open event. The project's strongest portfolio value is not simply that a campaign was launched, but that the environment was kept safe, evidence was preserved, and the results were interpreted cautiously.

Appendix A. Project Summary

IG-AWARENESS-01 is a localhost-only GoPhish test using MailHog as a local SMTP sink. It documents campaign setup, HTML email inspection, an educational landing page, and review of exported event and results data. The test uses a synthetic .test recipient and does not target real users or collect credentials.

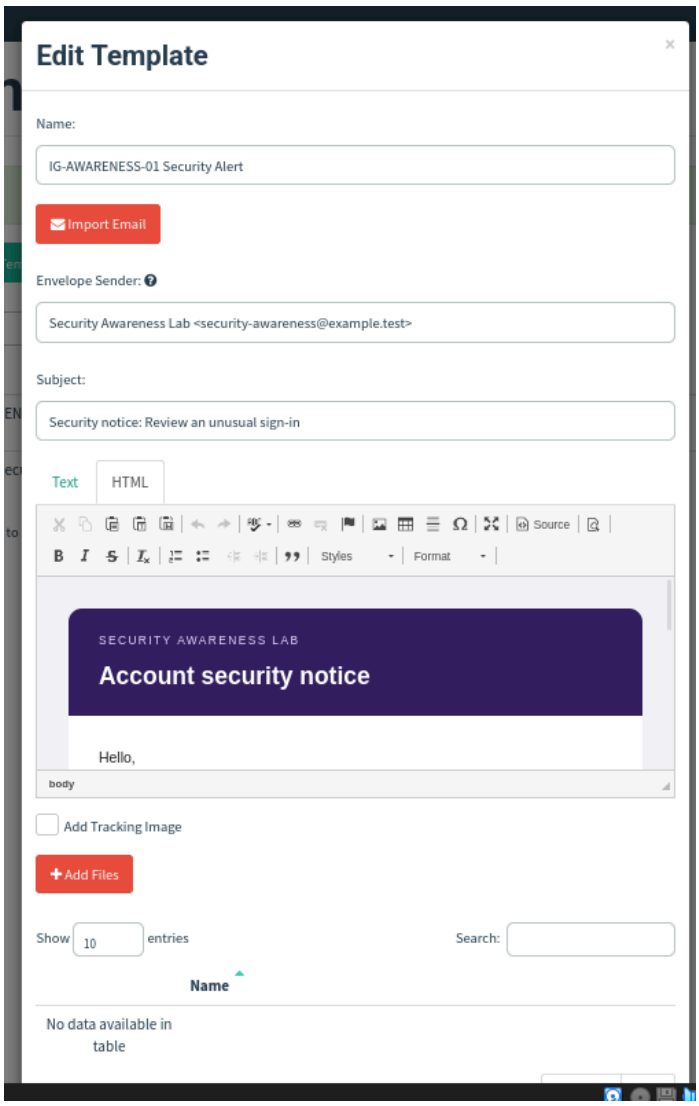
Appendix B. Publication and Evidence Checklist

- ☐ Replace [Your name] and any remaining personal-information placeholders.
- ☐ Check the document's page breaks, tables, and screenshot filenames.
- ☐ Keep the 11 evidence images in a clearly named folder.
- ☐ Include the two CSV exports as supporting evidence if appropriate.
- ☐ Do not publish GoPhish database files, session tokens, passwords, or private configuration.
- ☐ Export a PDF copy for easy review and keep the editable DOCX as the source document.

Appendix C. Screenshot Evidence

The following figures preserve the visual evidence collected during the local test. They document configuration and observed behavior; they do not establish that a real external recipient interacted with the message.

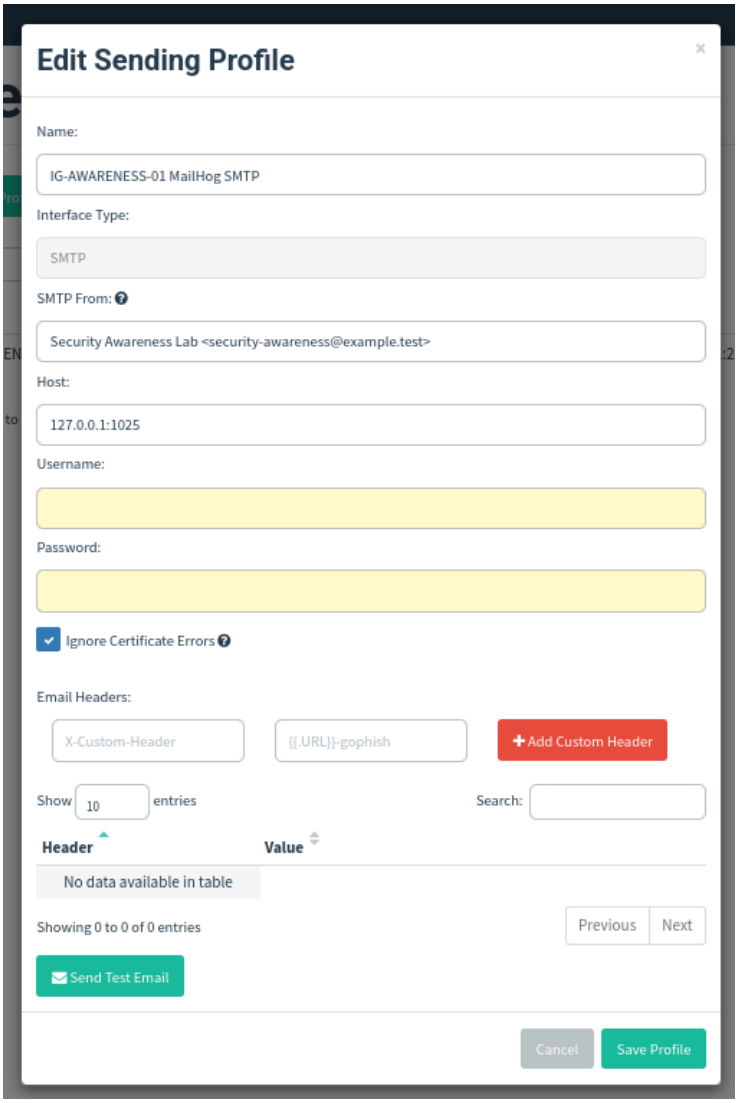
Figure 01. Email template preview



Shows the custom HTML security-awareness email template and subject/content design.

Evidence file: 01 Email Template Preview.png(1).png

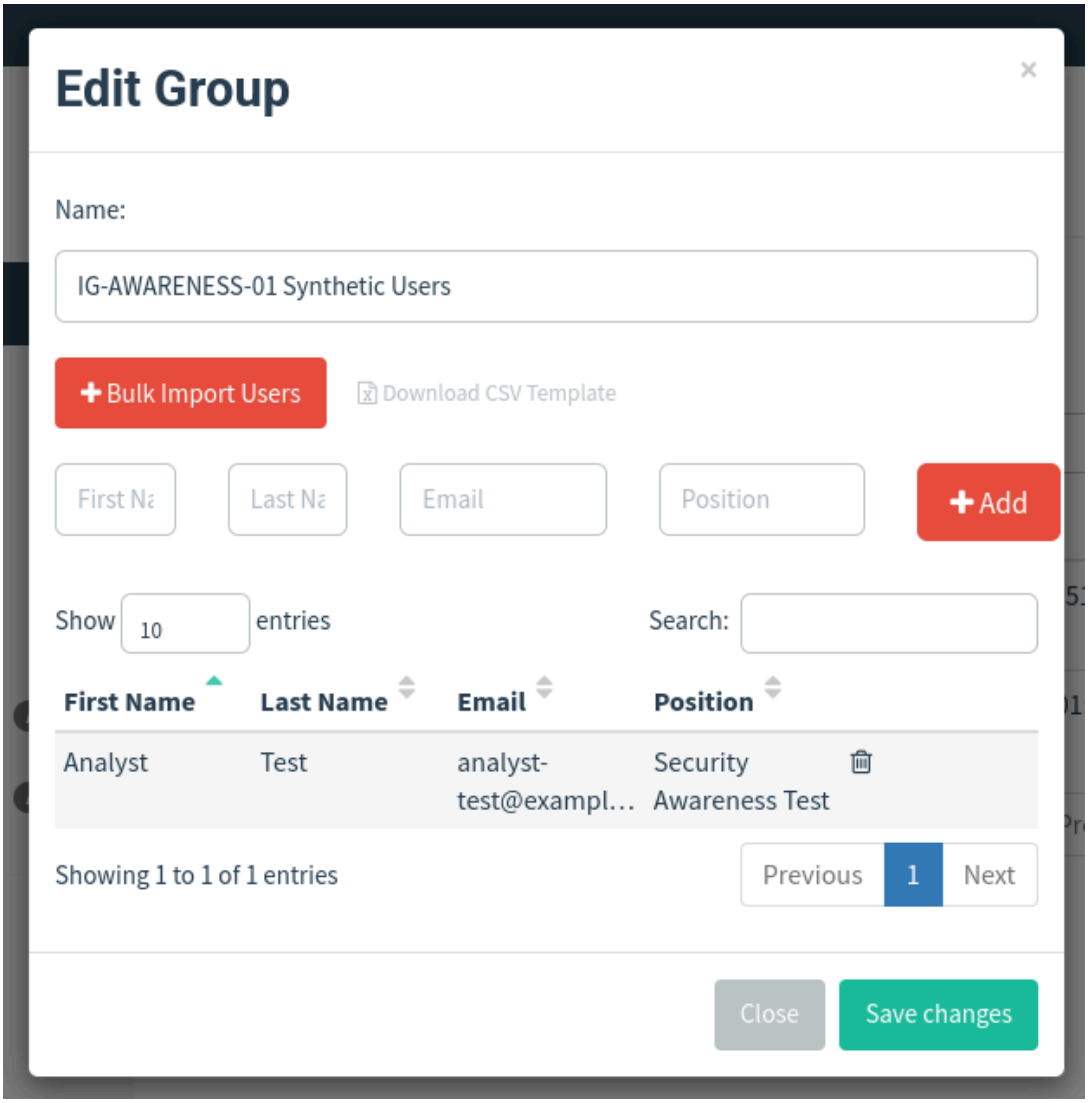
Figure 02. Sending profile



Shows the local MailHog SMTP sending profile configuration.

Evidence file: 02 Sending Profile.png.png

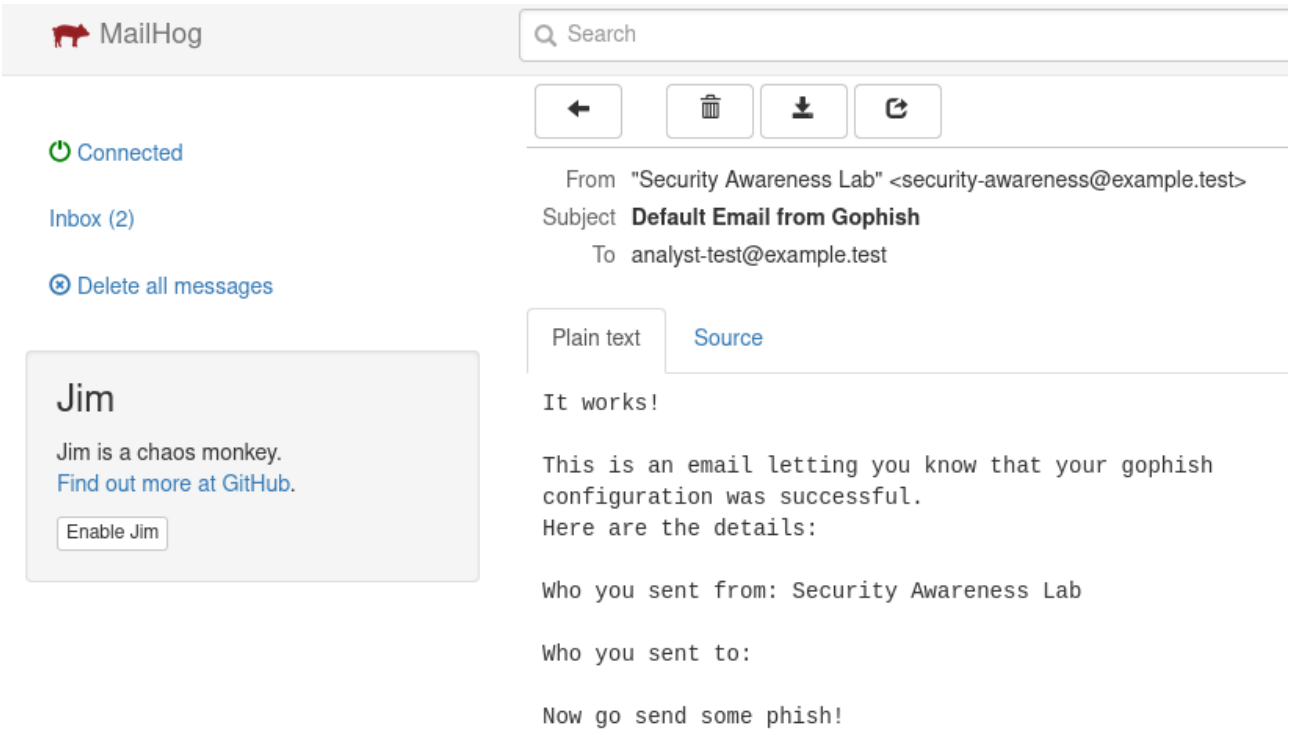
Figure 03. Synthetic recipient group



Shows the group containing the synthetic test recipient.

Evidence file: 03 Test Recipient Group.png.png

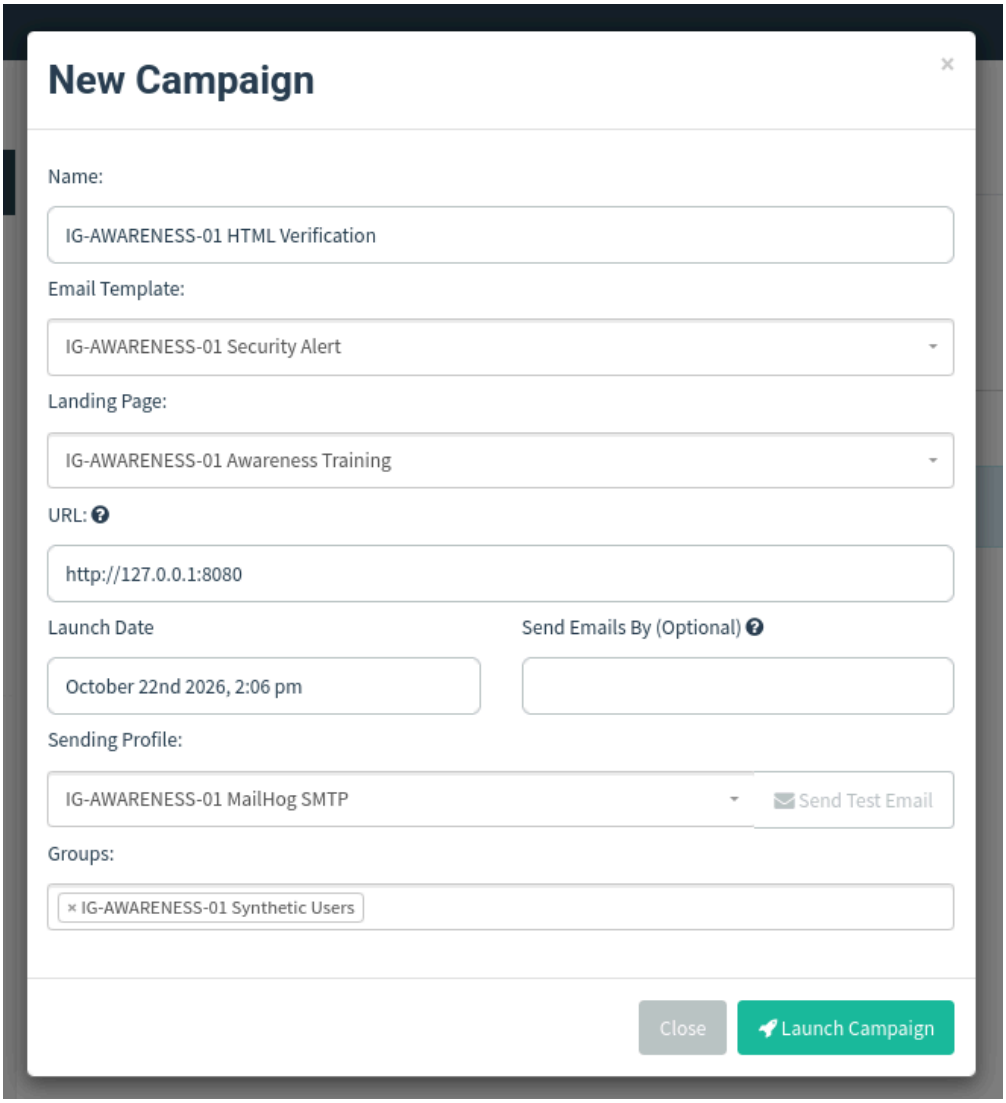
Figure 04. SMTP test email in MailHog



Shows GoPhish's default test email captured by MailHog. This verifies the SMTP test path; it is not the designed campaign email.

Evidence file: 04 SMTP Test Email.png

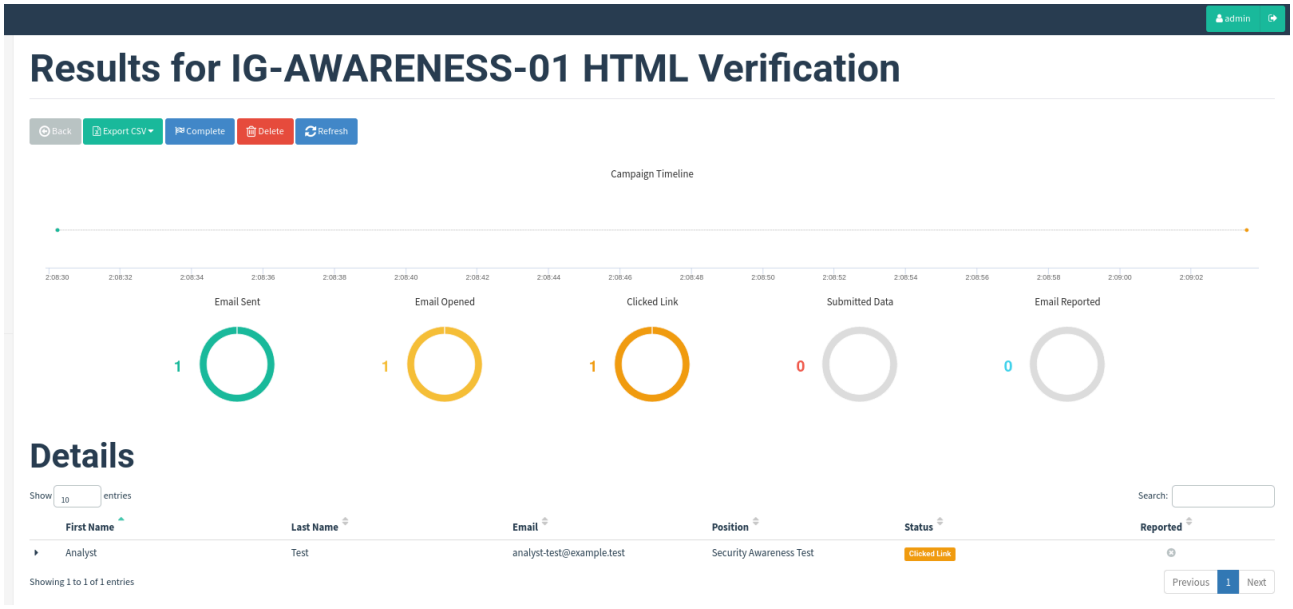
Figure 05. Campaign setup



Shows the campaign configuration, selected assets, local URL, sending profile, and test group.

Evidence file: 05 Campaign Setup.png.png

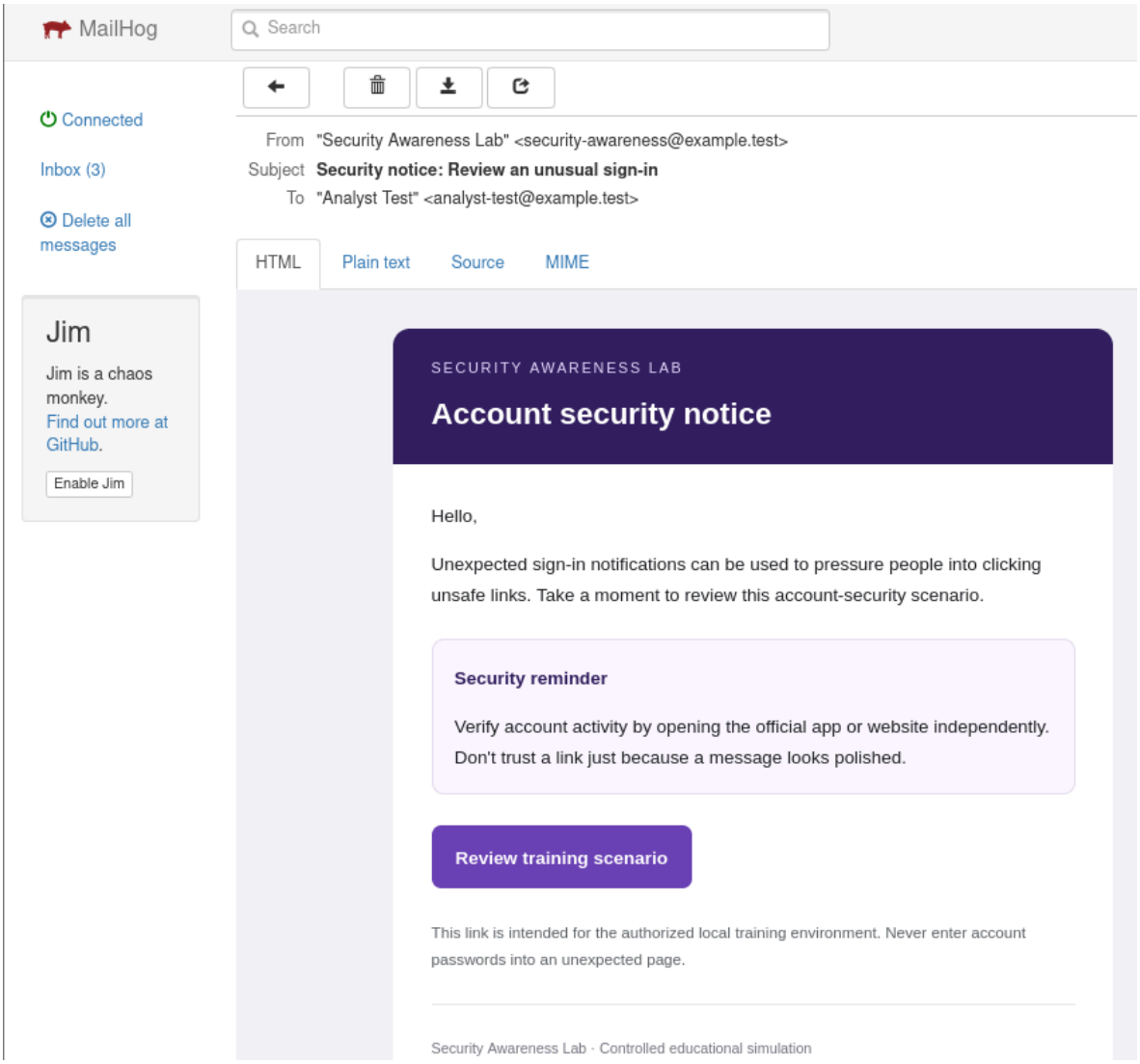
Figure 06. Campaign results dashboard



Shows the dashboard metrics captured during the controlled local simulation.

Evidence file: 06 Campaign Results.png.png

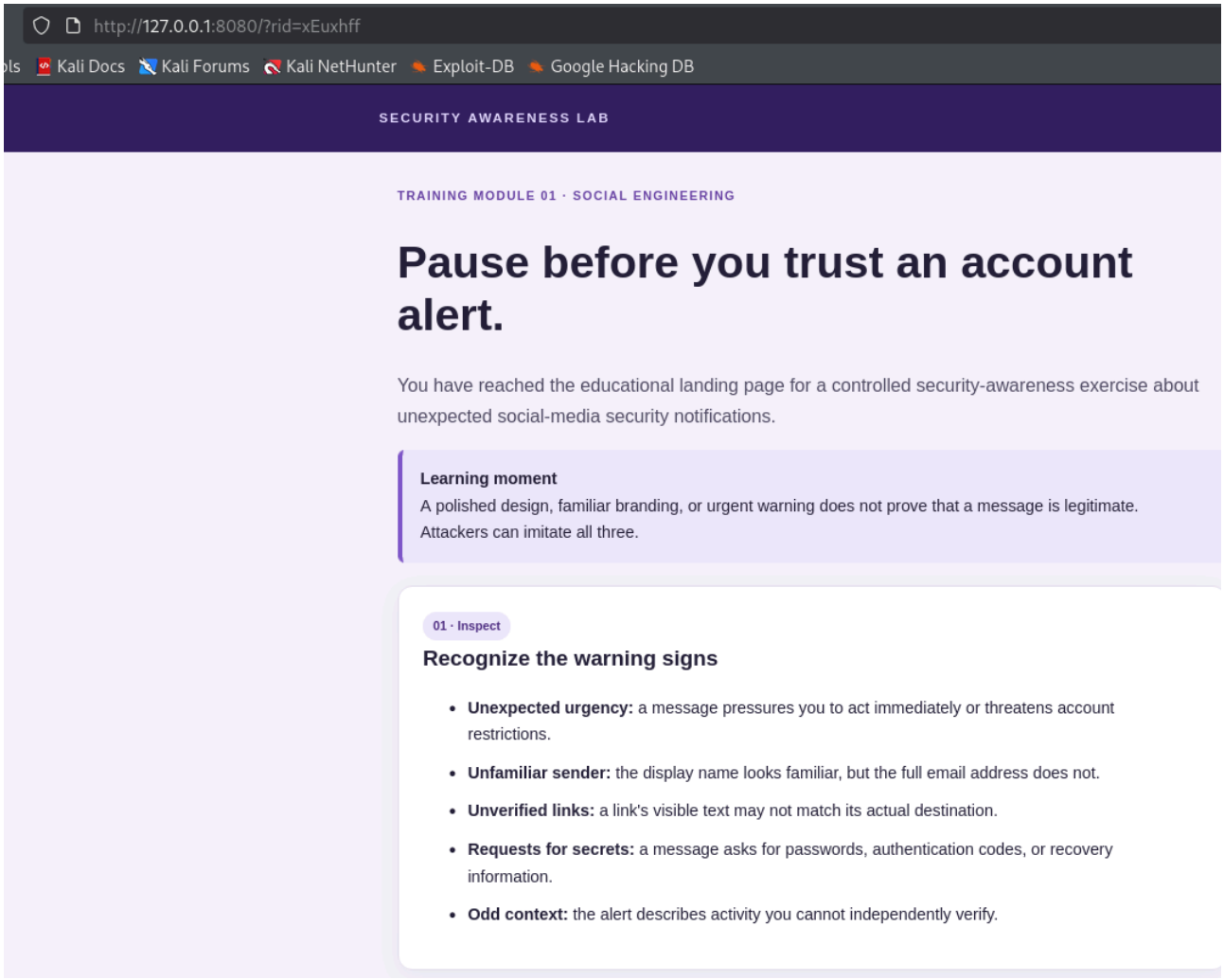
Figure 07. Campaign HTML email in MailHog



Shows the designed campaign email captured by MailHog.

Evidence file: 07 HTML Email in MailHog.png.png

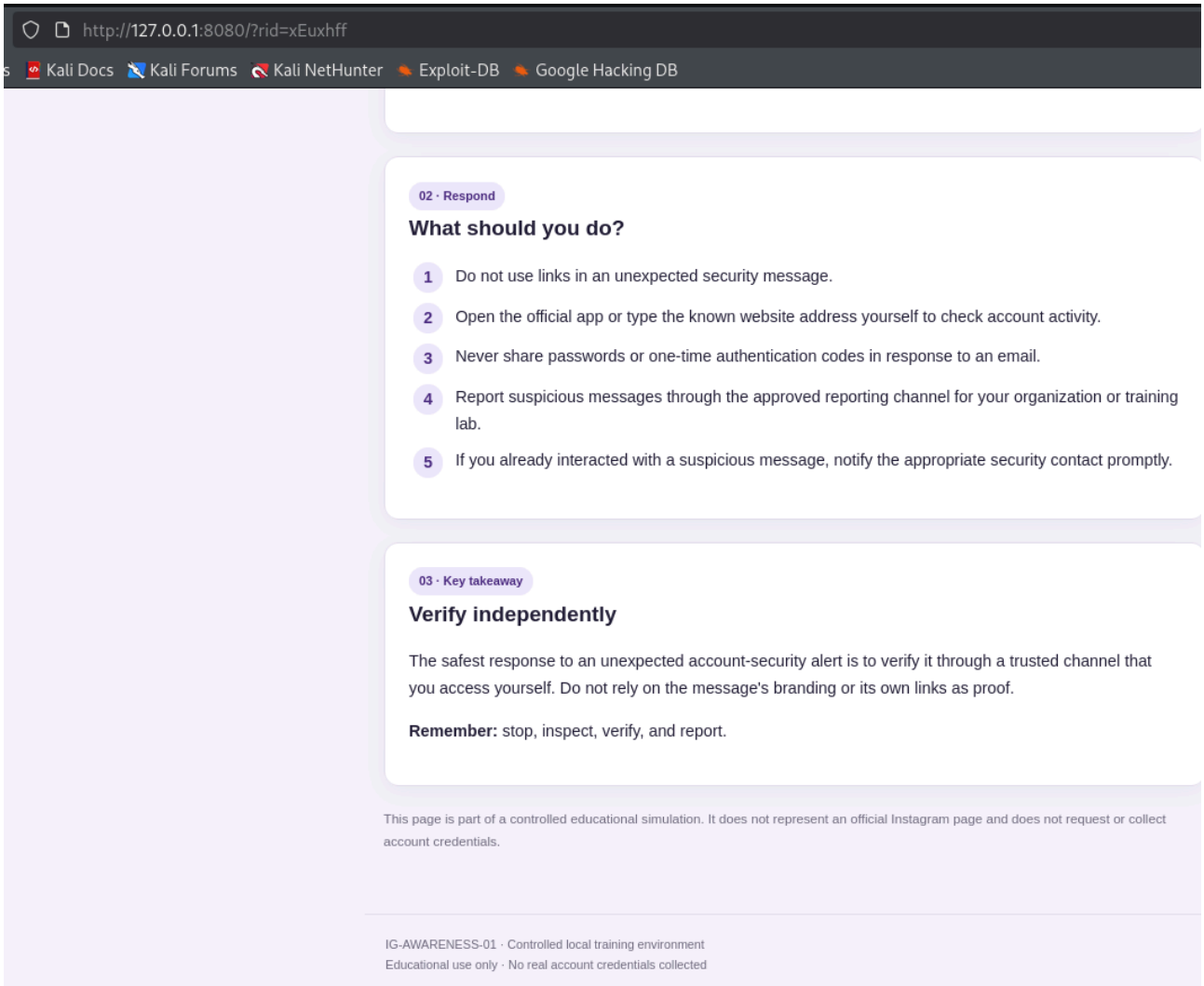
Figure 08. Educational landing page: top



Shows the landing page introduction and warning-sign section.

Evidence file: 08 Landing Page Preview - Top.png.png

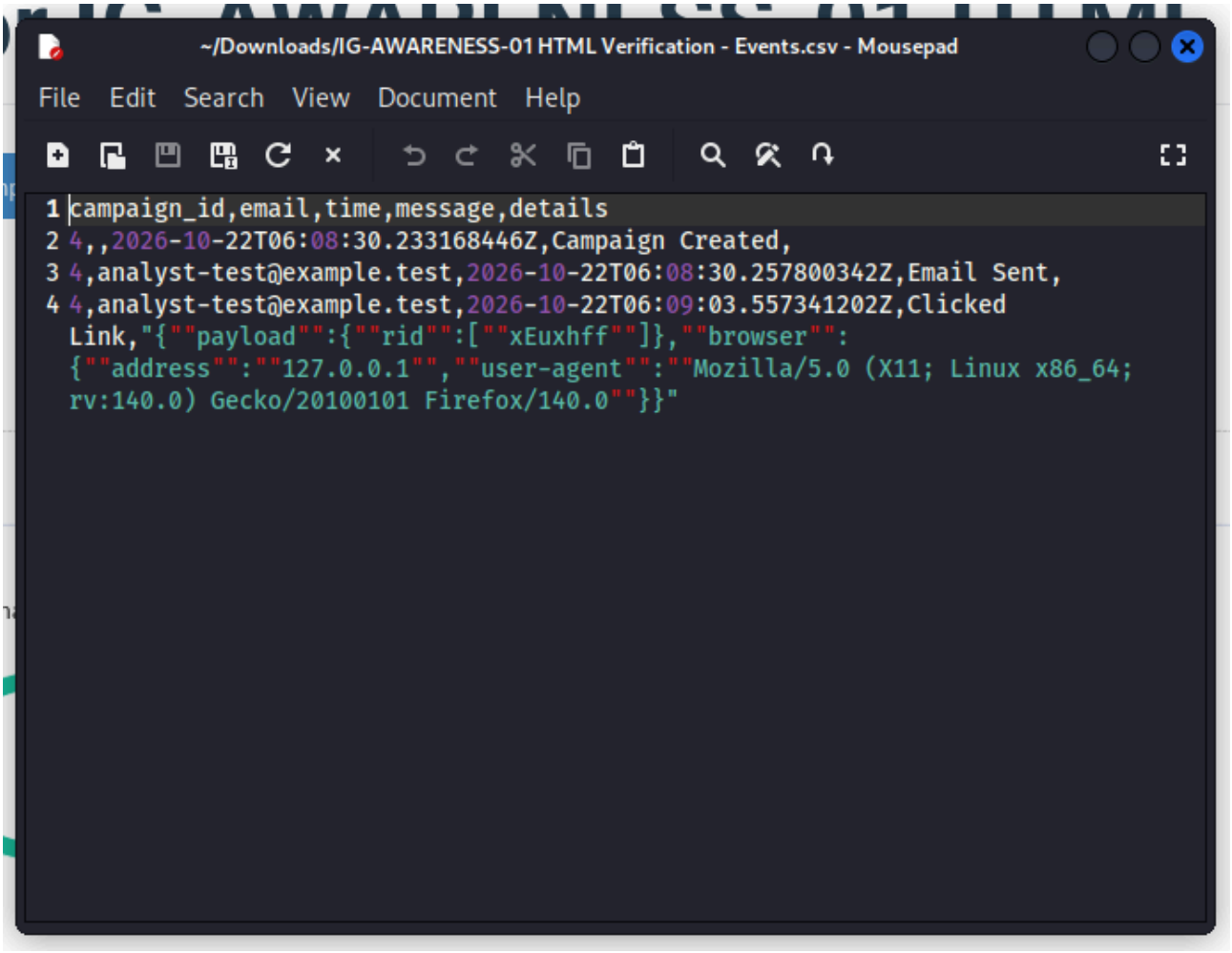
Figure 09. Educational landing page: bottom



Shows the response guidance, key takeaway, and educational-only notice.

Evidence file: 09 Landing Page Preview - Bottom.png.png

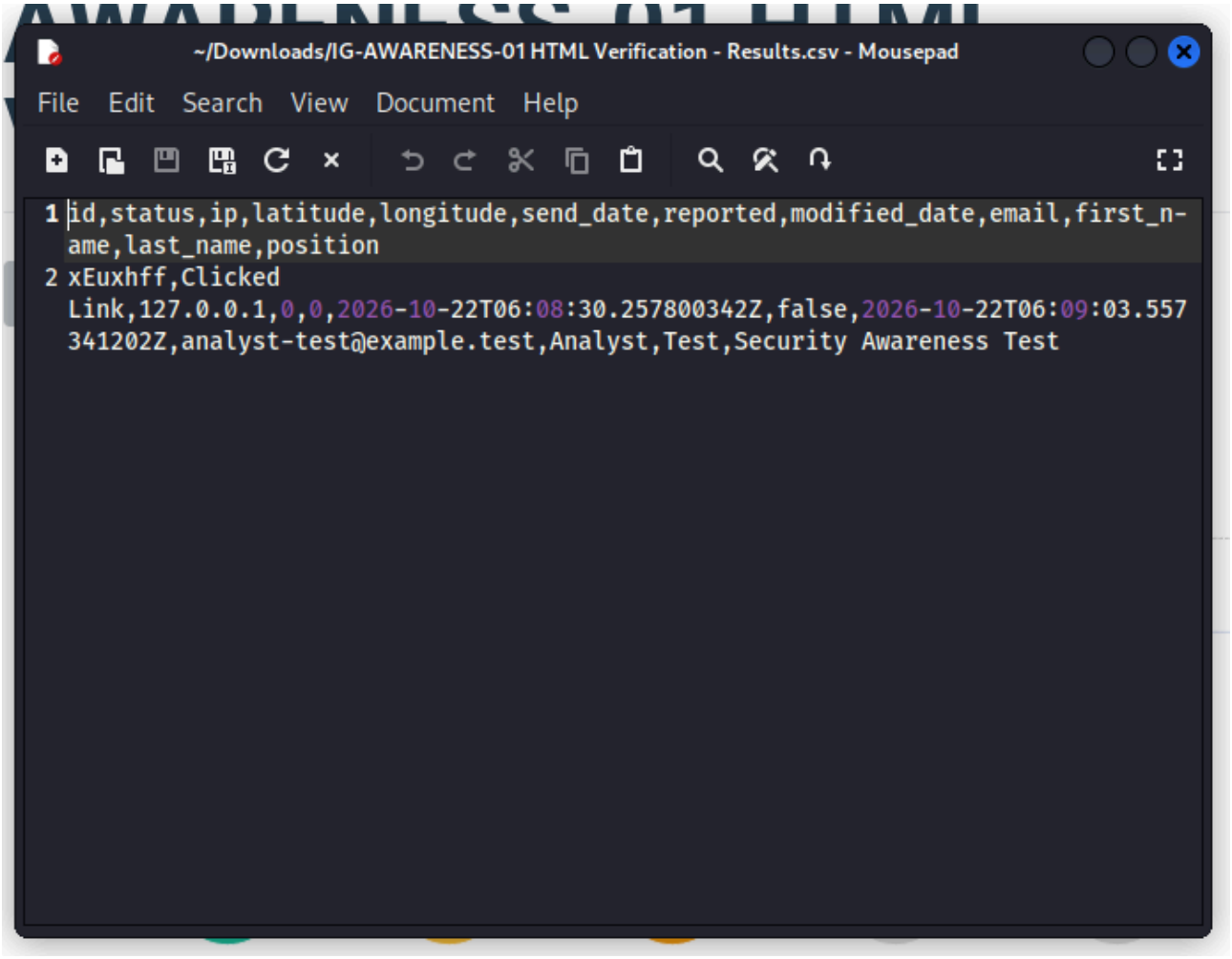
Figure 10. Raw campaign events CSV

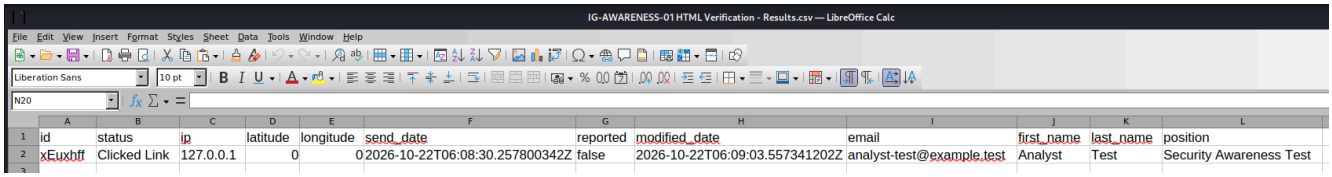


Shows the exported event records. The visible records include campaign creation, email sent, and link click.

Evidence file: 10 Campaign Raw Events.png.png

Figure 11. Campaign results CSV





Shows the exported recipient results, including the Clicked Link status.

Evidence file: 11 Campaign Results CSV.png.png